CS315 · ALGORITHMS ANALYSIS & DESIGN

Mock Midterm

A practice exam on Lectures 1 to 3 · 90 minutes · 25 marks

Muharib

01 Before you start

This exam is my own, in the format of the instructor's midterm (25 marks, one and a half hours). The questions are new and the numbers are different.

Important

This exam covers Lectures 1 to 3 only, because those are the ones I have. The real midterm also has questions from the later lectures (matrix chain, LCS and B-tree). Practise those from the past midterm files.

- Set a 90-minute timer.
- The solutions are in a separate file. Open it only after you finish.

02 Questions

Q1 HASH TABLE · 4 M

(A) Insert integer keys A = [5, 28, 19, 15, 20, 33] in order into a hash table of size 9 using the hash function h1(k) = k mod 9. Collisions should be resolved via chaining, where collisions are stored at the end of a chain. Fill in the hash table after all keys have been inserted. [2 pts]

(B) Collisions should be resolved via double hashing, and h2(k) = 1 + (k mod 5). Fill in the hash table after all keys have been inserted. [2 pts]

0	1	2	3	4	5	6	7	8

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Q2 LINEAR PROBING · 3 M

Insert keys 20, 33, 7, 46, 11, 59, 24 in order into a hash table of size 13 using h(x) = x mod 13. Collisions should be resolved via linear probing.

a) Fill in the hash table after all keys have been inserted. [2 pts]

b) How many probes did key 59 need? [1 pt]

0	1	2	3	4	5	6	7	8	9	10	11	12

Q3 COUNTING OPERATIONS · 3 M

Count the total primitive operations of the following algorithm in the worst case, then in the best case.

```
Algorithm arrayMin(A, n)
   currentMin ← A[0]
   Temp ← A[0]
   k ← 0
   for i ← 1 to n − 1 do
      if A[i] < currentMin then
         currentMin ← A[i]
   { increment counter i }
   return currentMin
```

Q4 TRUE / FALSE + MCQ · 5 M

Choose the correct answer:

1) Theoretical analysis allows us to evaluate the speed of an algorithm independent of the hardware/software environment. a. True b. False

2) Merge sort takes $O(n^2)$ time in the worst case. a. True b. False

3) With chaining, a colliding item is placed in a different cell of the table. a. True b. False

4) An algorithm which breaks a problem into sub-problems of the same form, solves them recursively and combines their solutions is: a. Greedy method b. Divide and conquer c. Dynamic programming d. None of the mentioned

5) The expected running time of findElement in a hash table is: a. O(1) b. O(log n) c. O(n)

Q5 BIG-OH · 3 M

a) Using the definition of Big-Oh, show that $6n^2 + 4n + 8$ is $O(n^2)$. Give c and n_0. [2 pts]

b) Order from the slowest growth to the fastest: n log n, 3n, n^2, log n, 2^n. [1 pt]

Q6 MERGE SORT · 4 M

a) Apply merge sort to the array [9, 3, 7, 1, 8, 2, 6, 4]. Show the divide steps and the merge steps. [2 pts]

b) Write the equation of T(n), the time to run MERGESORT on a list of size n. [1 pt]

c) For n = 32: how many levels does the recursion tree have, and how many problems are at level 2 and of what size? [1 pt]

Q7 SHORT ANSWERS · 3 M

a) Discuss all available methods for handling collisions in hash tables. [2 pts]

b) When can you say that the hash function is "good"? [1 pt]